



## TEST 2 FSPM0034 SEM3 1920

Statistics and Probability (Universiti Teknologi Malaysia)



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**FOUNDATION PROGRAMME**

**FSPM0034**

**Statistics and Probability**

**Semester 3 2019/2020**

**TEST 2(20%)**

**Duration: 90 minutes**

**INSTRUCTION: ANSWER ALL QUESTIONS AND SHOW ALL YOUR WORKINGS.**

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1. In a group of 12 persons, four are left-handed. Suppose that two persons are randomly selected from this group without replacement. Let  $X$  denote the number of left-handed persons. Determine the probability distribution of  $X$ . (4 marks)

2. A discrete random variable  $X$  has the probability function which is given as follows;

$$P(X = x) = \begin{cases} \frac{x+2}{20}, & x = 2, 3, 4 \\ kx, & x = 5, 6 \\ 0, & \text{otherwise.} \end{cases}$$

- i. Show that  $k = \frac{1}{44}$ . (2 marks)
  - ii. Hence, calculate  $P(3 \leq X < 6)$ . (2 marks)
  - iii. Determine  $E(X)$  and  $Var(X)$ . Hence evaluate  $Var(\sqrt{5}X - 2)$ . (4 marks)
3. A continuous random variable  $Y$  has the probability density function given by;

$$f(y) = \begin{cases} y; & 0 \leq y < 1 \\ c - y; & 1 \leq y < 2 \\ 0; & \text{otherwise} \end{cases}$$

where  $c$  is constant.

- i. Show that  $c = 2$ . (2 marks)
  - ii. Determine the cumulative distribution function for  $Y$ . (3 marks)
  - iii. Compute  $P(Y > 0.5)$ . (2 marks)
4. Let random variable  $X$  denote the time (in years) it takes to develop software. Suppose that  $X$  has the following distribution function;

$$F(x) = \begin{cases} 0; & x < 0 \\ x^5; & 0 \leq x \leq 1 \\ 1; & x > 1 \end{cases}$$

- i. Compute the probability that it takes more than three months to develop the software. (2 marks)
- ii. Compute the expected number of years it takes to develop software. (3 marks)

5. a) It is claimed that 45% of letters are delivered within two days of the time of mailing. Nine letters are randomly sent to different locations.
- What is the probability that exactly four letters will be delivered within two days? (3 marks)
  - Find the probability that from three to seven letters will be delivered within two days. (2 marks)
  - Find the expected number and variance of the letters that will be delivered within two days. (2 marks)
- b) The probability that a student from the Foundation program owns a car is  $p$ . A random sample of 12 students is selected.
- Given that  $p = 0.15$ , find the probability that at least seven students own a car. (2 marks)
  - Suppose the probability that none of the students own a car is 0.05, find the value of  $p$  to three decimal places. (2 marks)
6. The probability that a type of antibiotics can cure a certain disease is 0.85. If seven patients are given the antibiotics, find the probability that
- not less than four patients cured after taking the antibiotics. (2 marks)
  - between two and five patients do not cure after taken the antibiotics. (3 marks)

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## LIST OF FORMULA

### Random Variable

$$E(X) = \sum xP(X = x)$$

$$E(X) = \int xf(x)dx$$

$$Var(X) = E(X^2) - [E(X)]^2$$

### Probability distributions

| Distribution | Probability functions                                                                         | Mean   | Variance        |
|--------------|-----------------------------------------------------------------------------------------------|--------|-----------------|
| Binomial     | $P(X = x) = \binom{n}{x} \pi^x (1 - \pi)^{n-x}$ $x = 0, 1, \dots, n, \quad 0 \leq \pi \leq 1$ | $n\pi$ | $n\pi(1 - \pi)$ |